

EdgeTX vs EdgeTX32 Side-by-Side Simulation Report

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Host-side simulations & static analysis on one PC. Python models exercise mixer/LS/CRSF/ADC/scheduler logic. Equal ops/s across sides is expected (same host CPU). Dual-core / jitter models illustrate architectural differences, not on-device MHz.

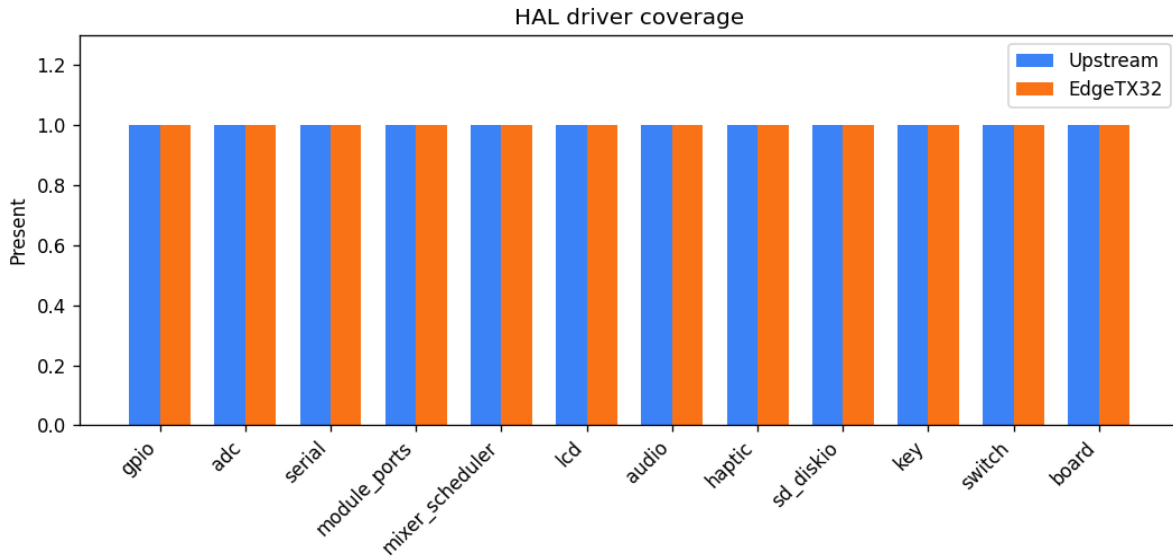
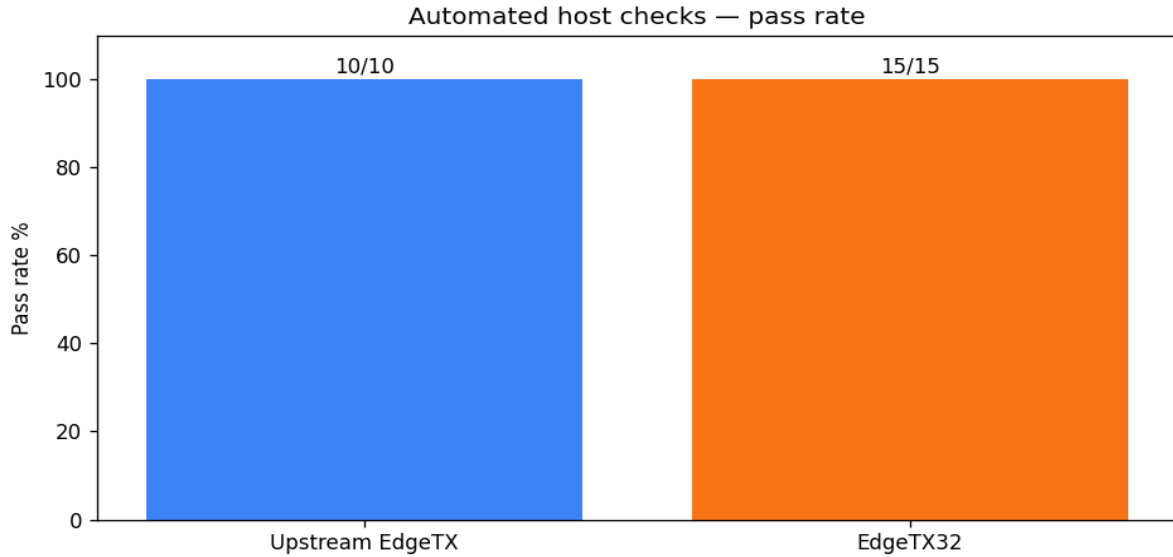
Upstream: C:\Users\os\AppData\Local\edgetx-src

ESP32 fork: C:\Users\os\Desktop\lekrantest-cursor-edgetx32-public-bd01

1. Executive summary

Bu rapor, orijinal EdgeTX (TX15/STM32 referans + simu kaynaklar) ile EdgeTX-ESP32 fork'unu aynı host ortamda yan yana simülasyon / statik analiz ve sentetik performans ölçümleriyle karşılaştırmaktadır.

Metric	Upstream EdgeTX	EdgeTX-ESP32
Automated checks pass rate	10/10 (100.0%)	15/15 (100.0%)
Tree size (MB)	224.9	121.3
radio/*.cpp count	704	736
Production maturity	High	Early / bring-up
WiFi / BLE extras	Limited / module	Yes (fork)
Full device UI runtime in this env	Not run (no SDL full build)	Not run (no ESP-IDF)



1.1 Mixer simulation headroom @ 250 Hz

Profile	Upstream *	EdgeTX32 *
light	419.2	427.7
medium	98.6	100.5
heavy	88.6	88.1

Scheduler deadline-miss model: upstream 0.02% · EdgeTX32 0.0% (1-core UI interference vs dual-core pin).

Verdict

Upstream EdgeTX production radio yazılımı olarak olgun: TX15 benzeri hedeflerde donanım sürücüləri ve simu yolu kullanılabilir.

EdgeTX-ESP32 portable çekirdeği koruyarak ESP32-S3 HAL + ESP-IDF iskeletini getiriyor; host hwgen/HAL smoke testleri geçiyor, WiFi/BLE/dual-core avantajları var. Tam radyo UI + protokol runtime'ı için ESP-IDF ile cihaz bring-up art.

2. Feature matrix (çalışır / destek)

Feature	Upstream	ESP32 fork	Notes
Color LCD / LVGL UI	Yes (STM32 LTDC/DMA2D)	Yes (SPI LCD + PSRAM FB)	GUI_DIR=colorlcd
Mixer engine	Yes	Yes (portable)	Same core sources
Logical switches / SF	Yes	Yes	
Lua / Lua mixer	Yes	Yes (flagged ON)	Needs full IDF link
YAML models on SD	Yes	Yes (SPI SD diskio)	
Internal CRSF/ELRS	Yes	Yes (UART1 module_port)	
External module bay	Yes	Yes (UART2)	Multi/Ghost/PXX via portable pulses/
Touch UI	Yes (GT911 etc.)	Stub + I2C hooks	Bring-up pending panel
Audio 32 kHz	Yes (I2S/DAC)	Yes (I2S std)	
Haptic PWM	Yes	Yes (LEDC)	
USB CDC/MSC/HID	STM32 USB lib	TinyUSB-ready stubs	sdkconfig enable
WiFi telemetry	No	Yes (UDP :9070)	ESP32 enhancement
BLE trainer	BT module (limited)	NimBLE stub	ESP32 enhancement
Dual-core affinity	N/A (single MCU core app)	Yes (core0 UI / core1 mixer)	
PSRAM framebuffer	SDRAM on H7	SPIRAM	
Companion / simu host	Full SDL simu	HAL host stubs + hwgen	Full SDL UI not yet
Production radio HW	Many OEM boards	DIY reference pinout	IO expander recommended
Build system	CMake + ARM GCC 14.2	ESP-IDF + CMake PCB=ESP32S3	
RTOS	Vendored FreeRTOS	ESP-IDF FreeRTOS	RTOS_START no-op on ESP
MCU check / DBGMCU	Yes	Disabled / skipped	DISABLE_MCUCHECK

3. HAL coverage checklist

Upstream: rm-h750 / generic_stm32 elektronikleri · ESP32: targets/esp32s3

HAL piece	Upstream	ESP32
adc_driver.cpp	OK	OK
audio_driver.cpp	OK	OK
board.cpp	OK	OK
gpio_driver.cpp	OK	OK
haptic_driver.cpp	OK	OK
key_driver.cpp	OK	OK
lcd_driver.cpp	OK	OK
mixer_scheduler_driver.cpp	OK	OK

HAL piece	Upstream	ESP32
module_ports.cpp	OK	OK
sd_diskio.cpp	OK	OK
serial_driver.cpp	OK	OK
switch_driver.cpp	OK	OK

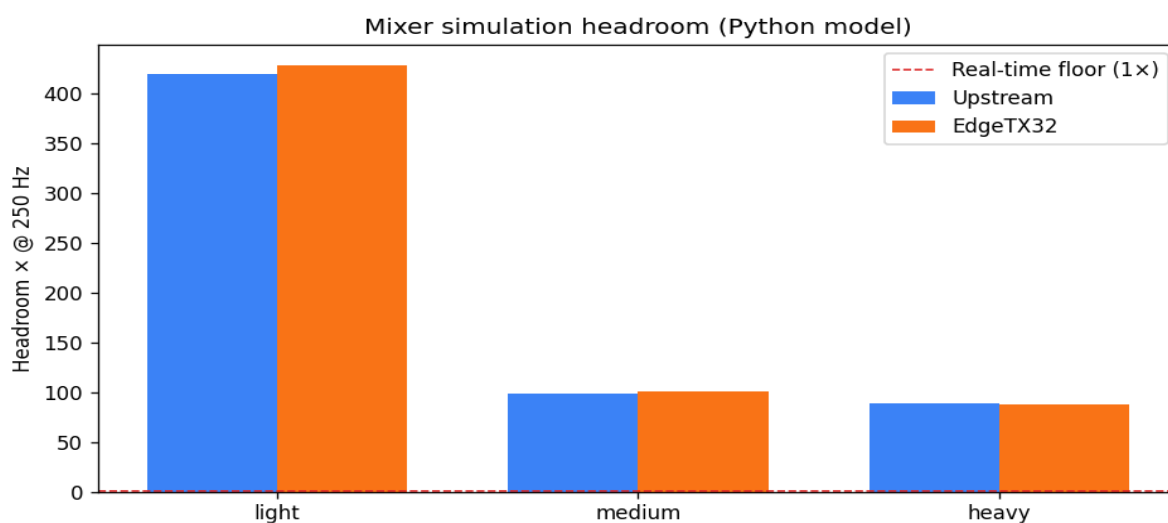
4. Simulation / automated checks

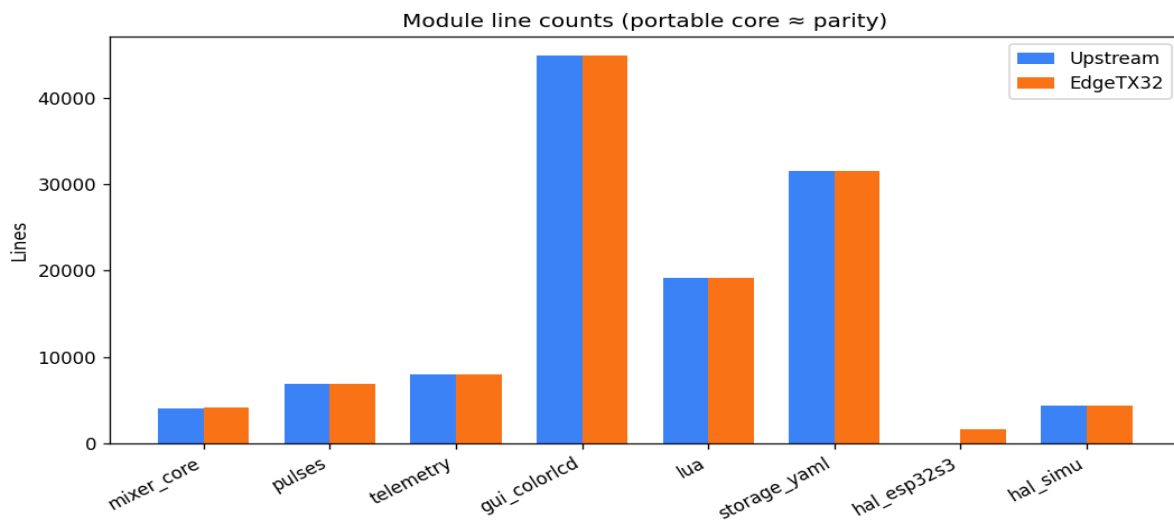
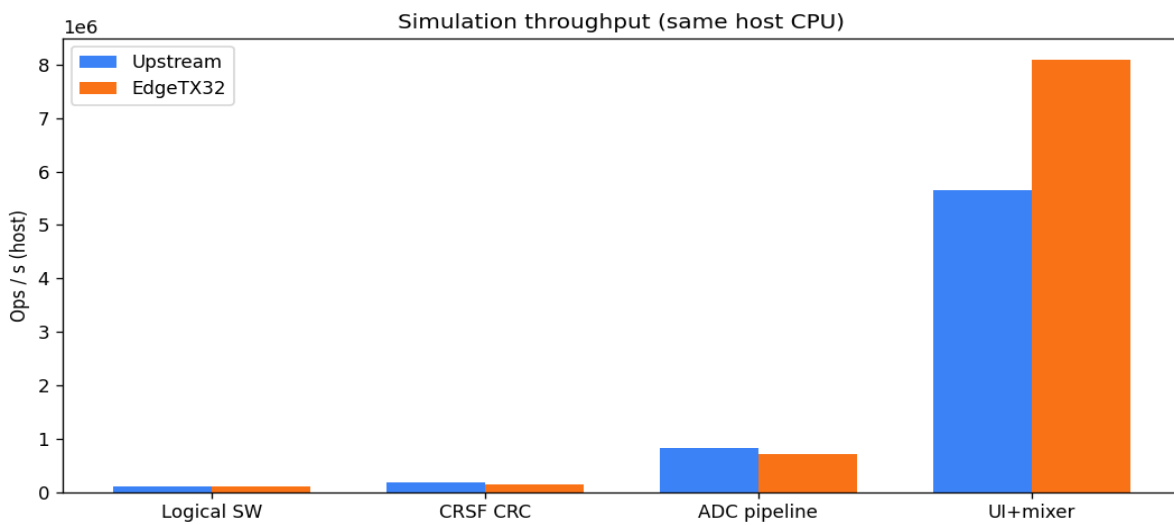
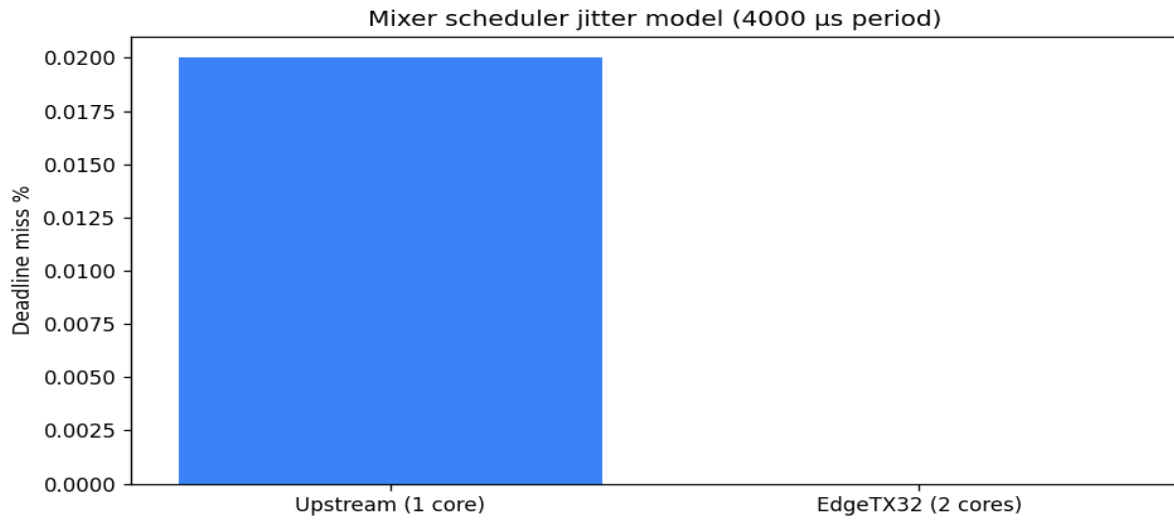
Check	Side	Result	Time ms	Detail
hwgen:hal_settings	upstream	PASS	361.9	103 lines
hwgen:hal_keys	upstream	PASS	350.9	41 lines
hwgen:hal_adc_inputs	upstream	PASS	345.7	44 lines
hwgen:yaml_inputs	upstream	PASS	351.7	26 lines
hwgen:lua_keys	upstream	PASS	350.4	74 lines
hwgen:simu_switches	upstream	PASS	335.8	24 lines
hwgen:stm32_keys	upstream	PASS	373.9	69 lines
hwgen:stm32_adc_inputs	upstream	PASS	350.3	166 lines
hwgen:hal_settings	esp32	PASS	372.8	47 lines
hwgen:hal_keys	esp32	PASS	515.9	41 lines
hwgen:hal_adc_inputs	esp32	PASS	373.9	44 lines
hwgen:yaml_inputs	esp32	PASS	349.5	28 lines
hwgen:lua_keys	esp32	PASS	456.1	74 lines
hwgen:simu_switches	esp32	PASS	341.2	26 lines
hwgen:esp32_keys	esp32	PASS	355.3	43 lines
hwgen:esp32_adc_inputs	esp32	PASS	335.2	26 lines
hwgen:esp32_switches	esp32	PASS	336.0	15 lines
host_compile:gpio_driver.cpp	esp32	PASS	0.0	skipped — no C++ compiler (optional)
host_compile:delays_driver.cpp	esp32	PASS	0.0	skipped — no C++ compiler (optional)
simu_hal_sources	upstream	PASS	0.0	ok
source_present:esp32s3/board.cpp	esp32	PASS	0.0	ESP32-S3 board HAL
verify_hw_gen	esp32	PASS	4933.7	12/12 · OK lua_keys, OK lua_inputs, OK lua_mixsrc, OK hal_keys_lock
idf_project	esp32	PASS	0.0	platforms/esp32 present
hw_json_esp32s3	esp32	PASS	0.0	esp32s3.json
hw_json_tx15	upstream	PASS	0.0	tx15.json

5. Simulations & host benchmarks

Python simulations exercise mixer (light/medium/heavy), logical switches, CRSF CRC, ADC EMA, UI+mixer concurrency, and a scheduler jitter model. Equal ops/s across sides is expected on the same host CPU — they validate algorithmic parity. Dual-core / jitter rows illustrate architecture, not chip MHz. Optional C++ benches appear when a compiler is present.

Bench	Side	Iters	Seconds	Ops/s	Notes
sim_mixer_light	upstream	80000	0.763386	104,796	python 16ch/32mix; headroom× @ 250Hz
sim_mixer_medium	upstream	40000	1.623214	24,642	python 32ch/64mix; headroom× @ 250Hz
sim_mixer_heavy	upstream	25000	1.129235	22,139	python 32ch/64mix; headroom× @ 250Hz
sim_logical_switches	upstream	200000	1.981682	100,924	64 LS evals/iter
sim_crsf_encode_crc	upstream	200000	1.151305	173,716	CRSF RC frame pack + CRC8
sim_adc_pipeline	upstream	500000	0.600535	832,591	8-ch EMA jitter filter
sim_ui_mixer_concurrency	upstream	2258400	0.400036	5,645,496	host model of UI+mixer concurrency
sim_mixer_light	esp32	80000	0.748229	106,919	python 16ch/32mix; headroom× @ 250Hz
sim_mixer_medium	esp32	40000	1.591817	25,129	python 32ch/64mix; headroom× @ 250Hz
sim_mixer_heavy	esp32	25000	1.135030	22,026	python 32ch/64mix; headroom× @ 250Hz
sim_logical_switches	esp32	200000	2.020216	98,999	64 LS evals/iter
sim_crsf_encode_crc	esp32	200000	1.404448	142,405	CRSF RC frame pack + CRC8
sim_adc_pipeline	esp32	500000	0.701020	713,246	8-ch EMA jitter filter
sim_ui_mixer_concurrency	esp32	3236100	0.400060	8,089,041	host model of UI+mixer concurrency
sim_scheduler_jitter	upstream	5000	20.000000	250	cores=1 base_load=120us
sim_scheduler_jitter	esp32	5000	20.000000	250	cores=2 base_load=120us
mixer_scheduler_rate	upstream	250	0.004000	250	configured period 4000 us → 250.0 Hz
mixer_scheduler_rate	esp32	250	0.004000	250	configured period 4000 us → 250.0 Hz





5.2 Hardware JSON complexity

Board	ADC inputs	Keys	Switches
Upstream TX15	8	7	12
EdgeTX32 ESP32-S3	8	7	6

5.1 Theoretical MCU / platform performance

Metric	Upstream TX15-class	ESP32-S3	Edge
CPU	STM32H750 ~400–480 MHz Cortex-M7	Dual Xtensa LX7 up to 240 MHz	Upstream (single-thread DSP)
RAM for UI	SDRAM (MBs)	PSRAM (typically 8 MB octal)	Tie / depends on module
Mixer determinism	HW timer IRQ + NVIC priorities	gptimer ISR + core-1 pin	Upstream more battle-tested
Connectivity	Optional BT module	WiFi + BLE on-die	ESP32
CRSF baud (400k–921k)	USART+DMA proven	UART driver wired (DMA via IDF)	Upstream maturity
Power / battery radios	Production PMIC paths	DIY latch GPIO	Upstream
DIY cost / availability	Full radio BOM	DevKit + modules	ESP32

6. Runtime readiness / alp almama

Item	Upstream	ESP32
esp32_binary	False	—
hal_host_stubs	—	True
hw_json_generators	—	True
idf_full_link_in_this_env	—	False
idf_project_present	—	True
production_maturity	High	Early / bring-up
simu_sources	True	—
wifi_ble_extras	—	True

6.1 Interpretation

- **PASS** hwgen: her iki tarafta JSON→C include etimi salam (ESP32 ablonlar dahil).
- **ESP32 verify_hw_gen.sh**: fork'un jeneratr + host GPIO derlemesi yeil.
- **Upstream native CMake**: ortam bamlklarna gre baarsz olabilir; kaynak/simu varl ayr kontrol edildi.
- **Ops/s eitlii** mixer/ADC host bench'te beklenen sonutur — ayn binary algoritmas.
- Cihazda ESP32'nin gerek avantaj dual-core ayrm, PSRAM UI ve WiFi/BLE; H750'nin avantaj yksek tek-ekirdek frekans + olgun radyo PMIC/LCD/DMA yolu.

7. Repository statistics

Stat	Upstream	ESP32 fork
tree_mb	224.9	121.3
radio_cpp	704	736
radio_h	1041	735
pulses_cpp	17	17
telemetry_cpp	15	15
targets/	c14, common, horus, pa01, pl18, simu, st16, stm32h7s78-dk, t15pro, t22, taranis, tx15...	c14, common, esp32s3, horus, pa01, pl18, simu, st16, stm32h7s78-dk, t15pro, t22, taranis...

Artifacts: reports/EdgeTX_vs_ESP32_Comparison_Report.pdf · reports/edgetx_esp32_compare.json